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PROFESSIONAL SUMMARY

ASIC Physical Design (PD) Engineer and Master of Technology graduate (VLSI Design) with hands-on exposure to **RTL-to-GDSII implementation flow** including **synthesis, floorplanning, placement, CTS, routing, and STA**. Proficient with Synopsys implementation and signoff tools including **Design Compiler (DC), IC Compiler II (ICC2), and PrimeTime**. Familiar with PD debugging via **Analyze and improve QoR metrics, physical sign-off activities including DRC/LVS and directed IR/EM analysis, congestion analysis, constraint tuning, and ECO-style iterations**. Strong foundation in digital design and hardware fundamentals with working knowledge of **UPF , lint/CDC concepts, Linux, and TCL**.

TECHNICAL SKILLS

- **ASIC Physical Design Flow:** RTL-to-GDSII, Synthesis, Floorplanning, Power Planning , Placement, CTS, Routing, Timing Closure (setup/hold), Congestion Analysis, Signoff report analysis, DRC/LVS, IR/EM analysis.
- **STA & Constraints:** SDC , STA analysis, path-based debug, clock tree optimization concepts.
- **Low Power & Checks :** UPF, linting/CDC awareness for robust implementation.
- **Languages/Scripting:** Verilog (RTL), SystemVerilog , TCL, Linux.
- **Synopsys Tools:** Design Compiler (DC), IC Compiler II (ICC2), PrimeTime, VCS, Verdi, SpyGlass.

TRAINING

ASIC Physical Design

Aug 2025 – Feb 2026

EDGE SENSE TECH SOLUTIONS Pvt. Ltd

- Performed physical implementation activities including floorplan creation, power planning, macro placement, placement optimization, congestion analysis, and QoR improvement for timing and routability.
- Conducted Static Timing Analysis (STA) for setup/hold analysis, clock optimization, timing violation debugging, and closure using PrimeTime with optimization techniques such as cell sizing, buffering, and constraint tuning.
- Worked on post-route analysis and physical signoff activities including timing, power, area analysis, DRC/LVS checks, IR drop, EM analysis, and developed TCL scripts for Physical Design automation.
- **Tools:** Design Compiler (DC), IC Compiler II (ICC2), PrimeTime, VCS, Verdi, SpyGlass.

PHYSICAL DESIGN PROJECTS

ASIC Physical Design of 32-bit Pipelined RISC-V Processor @ 32nm

Tools: Synopsys Design Compiler (DC), ICC2, PrimeTime

- Implemented complete physical design flow: synthesis, floorplanning, placement, CTS, routing, and signoff-oriented checks.

- Performed floorplan optimization through utilization tuning, placement, pin placement, blockages and keepouts to reduce routing congestion and improve timing QoR.
- Resolved congestion issues by using partial blockages and keepouts in hotspot areas.
- Applied useful-skew optimization to resolve setup/hold trade-offs on critical paths while monitoring clock latency and skew.

ORCA Core Processor – Physical Design Implementation

Tools: Synopsys ICC2, PrimeTime / instance count : 56k / Frequency: 433 MHz

- Executed full PD flow for ORCA core using ICC2 including floorplanning, placement, CTS, routing and QoR tracking.
- Manually placed and optimized macros to reduce blockages and improve routing channels; tuned utilization and placement to meet timing needs.
- Analyzed and fixed setup/hold timing violations using PrimeTime-driven debug and iterative optimization.
- Worked with a design size of ~56K standard cells and 40 macros.

Pipelined INT8 Systolic-Array AI Accelerator — RTL to GDSII Flow

28nm / instance count : 39k / Target Frequency: 500 MHz

- Designed and verified an 8×8 INT8 systolic-array accelerator with 64 MAC-based PEs for AI matrix-multiplication workloads. pipeline staging in the PE MAC datapath to reduce critical path delay and target 2 ns clock-period timing closure.
- Executed RTL simulation in VCS, synthesis in Design Compiler, and physical implementation in ICC2.
- Fixed timing and congestion violations during PNR stages, targeted positive setup/hold slack and implementation ready QoR.

EDUCATION

Master of Technology in VLSI System Design

OCT 2024 – AUG 2026

B V Raju Institute Of Technology (BVRIT), Narsapur, Telangana

- **CGPA:** 8.91 / 10.0
- **Relevant Coursework:** ASIC Design, Physical Design, Low Power VLSI Design, DFT, Digital VLSI Design, Advanced Physical Design.

Bachelor of Technology in Electronics and Communication

2020 – 2024

Kamala Institute of Technology and Science, Telangana

- **Honors/Leadership:** Elected Class Representative; Captain of the Cricket Team.

CERTIFICATIONS AND ACHIEVEMENTS

- **Synopsys Training:** Completed *RTL to GDSII Flow* workshop using Synopsys tool suite (Design Compiler, ICC2).
- **Physical Design:** VSD – Physical Design Flow (Udemy).
- **Verification (Foundation):** Design Verification with SystemVerilog & UVM (Udemy).
- **Programming:** C for Everyone: Programming Fundamentals (Duke University via Coursera).
- **Award:** Secured 1st Prize in *Technikwizz* (Technical Quiz), KITSOZONE 2023.